

Black-tailed prairie dogs look out from their burrow in a field in Wyoming. Study of this species has revealed its key role in fostering diversity in its native habitat.

Coral cleaners

The princess parrotfish in the Caribbean is another keystone species, this time because of the consequences of its feeding. The fish lives around coral reefs, where corals fight each other for light, nutrients, and space. The parrotfish scrapes the surfaces of the corals to remove layers of algal seaweed to eat. If the parrotfish did not do this, clumps of seaweed would grow on the corals, smothering as well as chemically damaging the reef. If the parrotfish was overfished or died out from disease, the health of the reefs would rapidly deteriorate.

Landscape managers

On African grasslands, elephants smash down small and medium-sized trees for food, helping maintain savanna as grassland and opening up new areas that were formerly woodland. This destructive behavior helps maintain the feeding habitat for grazing animals such as zebras, antelope, and wildebeest. It also indirectly helps the predators that hunt the grazers—including lions, cheetahs, and hyenas—and the smaller mammals that